THE STABILITY AND SIGNIFICANCE OF MAGNETISM IN SEDIMENTARY ROCKS*

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ABSTRACT

This study was undertaken to investigate whether sedimentary rocks can retain directions of magnetization through long intervals of geologic time. The method of investigation is to observe the angular relations of the directions of residual magnetization to the bedding in deformed structures, or in beds that were laid down in a random manner. Systematic relationships found indicate that the direction of magnetization in certain selected sedimentary rocks remains unchanged for at least 200 million years.

I—INTRODUCTION

General statement

The Department of Terrestrial Magnetism, of the Carnegie Institution of Washington, has supported investigations of the magnetic properties of sedimentary rocks since 1937. Some of the results show, for example, that the horizontal component of the permanent magnetization of Pleistocene varved clays of New England varies ± 30° through a series of beds that represent a few thousands of years of deposition [Johnson, Murphy, and Torreson, 1948; see 1 of "References" at end of paper]. Laboratory studies, and observations of the direction of magnetization (often referred to hereafter as the polarization or magnetization) of modern sediments, have led to the conclusion that a fine-grained sediment bearing finely-divided magnetic particles may receive a polarization at the time of deposition. In addition to factors that normally control the positions assumed by all particles on coming to rest on the bottom (hydrodynamic forces, bottom character, Brownian movement, colloidal phenomena, etc.), magnetic particles in an appropriate size-range are preferentially oriented by the earth's magnetic field to an extent necessary to give to the sediment a measurable polarization that is substantially coincident with the magnetic field prevailing at the site of deposition. It is thought probable that the varves and at least some (possibly many) sedimentary rocks received their initial magnetizations in this manner. Igneous rocks are not under consideration in this study.

It was recognized that if this pattern of primary magnetization could be shown

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to be permanent, a property of considerable interest would be at hand. It should be possible then to determine how the direction of the earth's magnetic field has varied during geologic time and, if the pattern were not obliterated by folding and thrusting, the property could be useful in the solution of structural problems.

Investigation of the permanence of these directions of magnetization is thus of interest. Tests of the stability are necessary for a number of reasons: A torque acts on all magnetic particles when their direction of magnetization does not coincide with the direction of the earth's field and an insecure bonding of the particles to their matrix might allow them to be turned; it is uncertain that the direction of magnetization of individual particles remains unchanged over long periods of time; sediments may have been remagnetized in a new direction after deposition; a primary polarization might be obliterated by growth, within the rock, of material having a polarization in a different direction; and the apparent direction of the polarization in a sample of finite size may be affected by deformation processes such as flow or shear. Any of these conditions affecting a sediment would change its polarization; the polarization would then be regarded as unstable.

During the past several decades, numerous measurements have been made of the directions of magnetizations in a great variety of rocks; see Haalck [2, 1942] for summary. These magnetizations do not always coincide with the earth's present field, and many rocks whose known deformation dates back millions of years are magnetized almost contrary to the earth's field. We may thus infer that magnetizations do not necessarily change extensively and rapidly, but, on the other hand, there is doubt whether the present magnetization was acquired by some process at a remote time or whether it represents an adjusted average between an initial value and the present direction of the earth's field.

Other than by inference, there is little information on the stability of polarizations in time. Vestine [3, 1947], for example, has calculated the likelihood that the magnetic particles may be rotated in their matrix. The results suggest that appreciable rotation is quite improbable. The theory of domain structure of ferromagnetic material proposed by Kittel [4, 1946] may perhaps be applicable to the small magnetic particles in sediments. If so, one would not expect their directions of magnetization to alter with time. Johnson, Murphy, and Torreson, in unpublished laboratory-tests, have found evidence that thermal effects and vibration, of the magnitude that might be expected in 15,000 years, should have little, if any, effect on the polarizations of Pleistocene varved clays from New England.

Purpose of investigation

The present study was undertaken to evaluate whether any sedimentary rocks are capable of retaining a direction of magnetization without change over long intervals of geologic time by studying internal evidence obtained from the rocks themselves. Observations will be described indicating that, under conditions which are not unusual, certain selected sedimentary rocks retain their magnetization over very long periods.

Method of investigation

In this study, conclusions about the stability of polarizations are based on the patterns of magnetization observed in beds that were deformed at remote times.

The conclusions do not depend for their validity on assumptions regarding the chemical or physical history of the sediments, but on measurements which show that the deformation does not influence the initial direction of magnetization with respect to reference axes fixed within small-volume elements of the rock, and under appropriate circumstances (no shear or flow) this characteristic is exhibited by relatively large samples, showing up as a systematic relationship between the attitude of the bedding and the directions of polarization.

It is assumed that if polarizations change in time (except during periods of active deformation of the bed), they will change in the deformed bed in directions that will bring them into greater parallelism with one another and with either the earth's present magnetic field or with some other magnetic field that prevailed when the polarizations were acting in an unstable manner. Presumably the polarizations more oblique to the prevailing field will change more rapidly and more extensively than those closely aligned to the field.

Stability is a relative matter only. To say that a polarization has remained stable for a long time only expresses the fact that forces too weak to cause the polarizations to change have acted on them during the interval that they have been stable in direction. Polarizations in any material become unstable when certain conditions of heat, vibration, stress, or magnetic fields are applied. The significance of the observations presented below is that sedimentary rocks are found which have not been acted on by forces large enough to change their directions of polarization during the time which has elapsed since they were folded, in some cases as long ago as the Permian.

The most satisfactory deformation for the study of the stability of polarizations is a fold. Very frequently the manner in which various portions of a bed have been inclined to one another can be ascertained readily and the necessary rotations can be determined for restoring the bed point by point to the horizontal position. If the polarizations at various points in the fold are rotated correspondingly, they will again become parallel to one another if they have rmained unchanged since before the time of deformation, provided the process of folding tilted the polarizations in the same way as it tilted the bed, and provided the beds were uniformly magnetized initially. If, however, the polarizations diverge from one another after the restoration of the beds, then it must be concluded either that the folding process changed the polarizations or that the polarizations have changed since the time of deformation.

If a fold is to be useful for stability investigations, the initial direction of magnetization must not parallel the axis of folding, since the folding process would then not be expected to deflect the polarizations. In this study, one case will be described in which this situation was encountered, and several other cases will be presented which showed suitable relations between axis of folding and polarization to give information on the stability. A further case will also be treated in which folding evidently did not deflect the polarizations in the same way as it tilted the beds.

Another suitable geologic situation for stability studies is one in which random polarizations have been brought about, as in the case of a conglomerate. If the polarizations of all the pebbles are parallel, it may be concluded that the original polarizations in them no longer exist. If, however, the directions of magnetization

of the pebbles are random, the polarizations have remained, at least in part, unchanged since the pebbles were deposited. When the source bed of the pebbles is near-by, inferences about the stability of its polarization can be based on the observations in the conglomerate.

Sedimentary rocks may receive random polarizations during deposition. If, for example, the magnetic particles are too large to be turned into a preferential orientation by the earth's field, or if the conditions of settling are such that the orienting influence of the earth's field is overpowered by movements such as turbulence or stream flow, then the sediment may acquire a polarization which has no apparent relationship to the earth's field that existed during deposition. Observed randomness, especially in very small volume, may be taken as an encouraging indication of the stability of polarizations, but it is not yet subject to the same exactness of interpretation as the two geologic situations mentioned above.

Extent of work

Field and laboratory work extending over almost two years was required to obtain the data reported here. After formulating the problem and the mode of attack by studying folds, the writer spent about nine months participating in the development of instruments and technique, and another three months on a transcontinental expedition studying rock magnetism. On this expedition, few suitable opportunities were encountered for studying stability, but subsequent field-studies in the near-by Appalachians have yielded examples of the desired types of geologic situations.

Studies were initiated, in the order indicated, at the following localities, but were abandoned for the reasons given:

(*a*) Folded Paleozoic rocks of the Hudson River Valley: Apparatus not yet sufficiently sensitive to measure the weak moments
(*b*) Folded Cambrian quartzite at Harpers Ferry, West Virginia: Sampling difficulties
(*c*) Folded Cretaceous rocks near Walsenburg, Colorado: Too weakly polarized; sampling difficulties
(*d*) Folded Mesozoic rocks, San Rafael Swell, Green River Desert, Utah: Sampling difficulties
(*e*) Disturbed Eocene rocks, Green River, Wyoming: Igneous rocks mistaken for disturbed sediments
(*f*) Folded Cretaceous rocks near Garrison, Montana: Too weakly polarized
(*g*) Folded Cretaceous rocks 20 miles southeast of White Sulphur Springs, Montana: Too weakly polarized
(*h*) Folded Pre-Cambrian rocks 18 miles east of White Sulphur Springs, Montana: Too weakly polarized

II—TECHNIQUES EMPLOYED

Sample preparation

A variety of techniques have been used in preparing oriented rock-samples for measurement and the ones that have proved the most satisfactory are described below.

All orientations were measured to the nearest degree with a Brunton type pocket-transit. A small dural frame attached to it gives long straight-edges that improve the accuracy of measurements and orientation marks on samples.

On the bed to be sampled, a horizontal line is drawn with a diamond pencil and one end is marked with an arrow. Its azimuth (strike) is determined, and the dip line also is marked and its inclination noted. The slab with the orientation marks is then removed from the outcrop and preparatory to the coring operation it is blocked-up on a level surface so that both orientation lines lie in a horizontal plane. The original upper surface is not always placed uppermost in the coring operation, in order to avoid any systematic relationship to the coring apparatus, which is not entirely non-magnetic. Using short lengths of oiled boards, a crib is formed about the block and quick-setting plaster is poured around it, leaving the top surface with the orientation marks exposed. To minimize the chance of losing the orientation marks during the coring operation, a number of lines are drawn parallel to the strike line on the surface of the block. A 1-5/8 inch core, 1-1/2 to 2 inches long, is then cut from the block with a thin-walled bit having a cutting edge formed of diamond dust imbedded in a sintered metal matrix. Again the azimuthal orientation is random with respect to the coring equipment. Plain water, or water with soluble oil, under moderate pressure, is used for cooling and clearing the bit. "Dimet" bits, produced by Felker Manufacturing Company, Torrance, California, proved very satisfactory. A 3-cm cube is then cut from the core using brass core-holding jigs and an arrangement of 8-inch diamond saws. The edges of the cube are cut parallel to the two orientation marks and normal to the orienting surface. The magnetic measurements are carried out relative to these edges.

The procedure for preparing oriented cubes of soft rocks that cannot be cored still leaves much to be desired in the way of speed and convenience. A flat surface is scraped on the bed, orientation marks are scribed on it, the block is removed, and then using a miter-box, hack-saws, flat files, and sandpaper the 3-cm cube is fashioned as well as possible. Even with this admittedly primitive technique, cross-checks on samples have been found to be reasonably self-consistent and the results are considered satisfactory for our present purposes.

For sampling the soft varved clays, sample-taking cups 1-1/2 inches deep were prepared from 1-1/2 inch I.D. bakelite bonded-paper tubing. Bottoms of "Lucite," having a scribe line for reference, were cemented on with bakelite cement. After slight drying, the clay cores could easily be shaken out of the cup and then cubes cut using copper circular saws and special brass holding-jigs.

The resistance of the rock magnetism to change under the influence of the various sampling-procedures has been unexpectedly satisfactory. In many independent tests, no evidence has been found that the techniques employed change the state of magnetization of rocks. These tests have included drilling samples from a number of different directions with both the steel-walled bit and a special all-bronze bit, hitting the samples with hammers, and cleaning with quartz sandpaper the surfaces that might have been contaminated with steel from the tools. Also, it has been determined that the coring and cubing procedures do not magnetize rocks which are initially non-polarized, even though the same samples can be given strong residual magnetizations by applying momentarily a magnetic field of 100 gauss.

Measuring equipment

The measuring equipment was developed in 1947 at the Department of Terrestrial Magnetism as a refinement of an earlier apparatus by Johnson [5, 1938], built for studying weakly magnetized materials. It will be fully described by E. A. Johnson, Thomas Murphy, and P. F. Michelsen in a forthcoming publication. It operates on the principle that the faint residual magnetism of a rock is sufficient to induce an alternating voltage in a signal pick-up coil when the specimen is rotated close to the coil. This voltage is electronically amplified to a usable level, and from instantaneous comparisons of the phase of the voltage with the orientation of the sample relative to the pick-up coil, it is possible to determine the orientation in the rock sample of a plane, parallel to the axis of rotation, that lies parallel to the line through the north and south poles of the sample. To determine the direction of the north-south line, it is necessary to revolve the sample on at least two non-parallel axes and find the direction of intersection of the determined planes which lie parallel to the north-south line. Individual determinations of the directions of the planes usually require about one minute and have an accuracy of from $\pm$ 0°.5 to $\pm$ 5°, depending on the intensity of magnetization. In addition to determining the direction of magnetization, it is possible to determine the magnetic north and south sides of the sample. In practice, two measurements are made on each of the three axes. From these six measurements the direction of residual polarization in each sample is deduced, with a cone of error usually not exceeding 1° in radius. The intensity of magnetization is derived, with an absolute accuracy of about 5 per cent, from the observed values of the voltages produced by spinning the sample on different axes.

The minimum magnetic moment that can be detected when using a 3-cm cubic sample is 2×10^{-8} cgs/cc, but measurements of the planes are difficult and inaccurate at this intensity. With a moment greater than 3×10^{-6} cgs/cc, measurements are rapid and accurate.

Although the equipment is stable in operation, two calibrations are made. The intensity is calibrated by standard magnets which are checked occasionally by independent methods. Calibration of phase is made with each sample on the basis of symmetry considerations.

Field equipment

For field work, the measuring apparatus, together with a 3-KW portable electric generator and equipment for cutting oriented specimens, is mounted in a 2-1/2 ton, six-wheeled truck. Another similar truck supplies living facilities, and it is thus possible to carry out all measurements of rock magnetism in areas at some distance from permanently-established facilities. This arrangement is excellent for field work, but has the disadvantage that, owing to the slightly-variable line-frequency of the portable generator, the determinations of the direction in space of the magnetization can be in error by 1° to 5°, depending on the intensity of magnetization. Where well-regulated commercial power is available, it is used, and samples measured by using the portable generator are checked when operating on a well-regulated supply.

All data, with the exception of those from the varved clays, were obtained from samples which had been out of the outcrop less than four days. The varved clays were cubed and measured after a period of three months.

Thus, with the exception of the varves, these data are not subject to the pronounced changes in direction of magnetization that have been observed; for example, by Schneider, Thellier, and Thellier [6, p. 29, 1948] in samples prepared for measurement. Such recognized instability is not understood and has not been investigated by us, but may perhaps be related to the demagnetizing forces any magnet exerts on itself. The demagnetizing effect increases as the size of the magnet is decreased, and this effect is not large for material *in situ*. Our samples are parts of a much larger, uniformly-magnetized environment.

Solution of solid geometry

The Schmidt [7, 1925] equal-area projection is used as a rapid graphic means of solving the problems in solid geometry which are routine in this work. Its uses in many problems of geology have been discussed by Knopf and Ingerson [8, pp. 226–262, 1938] and Fairbairn [9, Chap. X, 1942]. In Appendix *A*, a brief discussion of the projection is given and its method of use with magnetic measurements illustrated.

Methods of presenting the data

With the exception of two plots on rectangular coordinates, all the data are presented as points in the lower hemisphere of the Schmidt equal-area projection. In all of these projections, the cardinal points on the horizon are indicated, and the nadir is shown by a cross in the center of the projection circle.

III—OBSERVATIONS

Pleistocene varved clays

General description—Field work was carried out during four weeks in August 1947 in the Pleistocene varved clays in the pit of the Michael Kane Brick Company, Middletown, Connecticut. The clays are exposed to a depth of about 20 feet along an east-west working-face several hundred feet long. Their areal extent is unknown and they occur on a slight rise above the surrounding country.

The deposit consists of layers of buff-yellow, fine, silty clay, 0.5 to 8 inches thick, alternating with layers of moist, generally impervious, dark milk-chocolate brown, greasy clay 0.5 to 3 inches thick. There is no evidence of induration or weathering in the section prepared for sampling, but surfaces which had been exposed for as little as two months contained characteristic concretions.

The clays were deformed during the Pleistocene shortly after their deposition, as demonstrated by Antevs [10, p. 117, 1928] and confirmed locally. Throughout most of the pit, the bedding, although horizontal on the average, shows gentle changes in dip, and in three places 200 feet apart flexure-fold anticlines involving all the beds exposed have been developed. Some of the crests of the folds are truncated and covered by a few feet of fine glacial till. Two reverse faults, transecting all the beds at about 60° with a few inches displacement, were noted.

Surfaces of discontinuity, best referred to as a form of fracture cleavage, making large angles with the bedding and acting apparently as planes of weakness across it, are a distinctive feature of the clay layers. They can be seen in a fresh exposure only after the surface has been very carefully scraped smooth, as closely-spaced

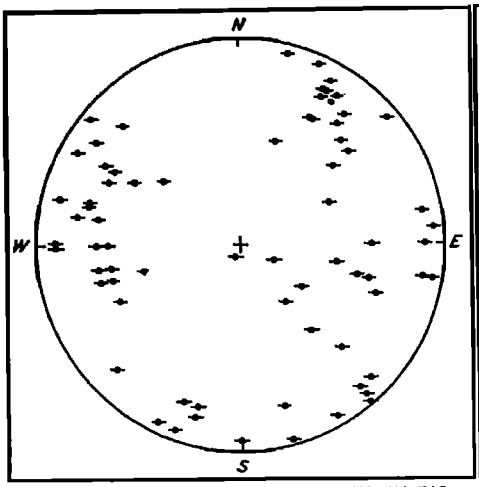


FIG. 1—POLES OF CLEAVAGE IN PLEISTOCENE VARVED CLAY PIT; MIDDLETOWN, CONN., SEPTEMBER, 1947

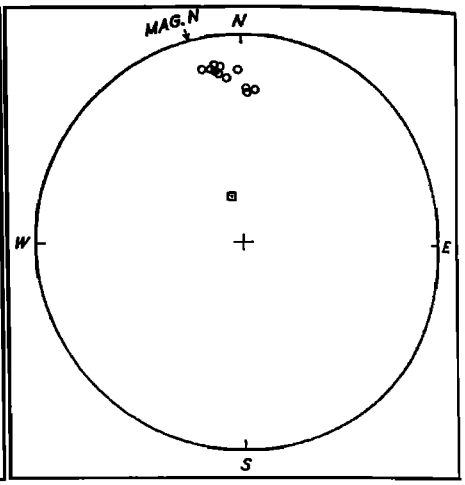


FIG. 2—SCATTER OF POLARIZATIONS IN A FLAT-LYING PLEISTOCENE VARVE; MIDDLETOWN, CONN., DECEMBER, 1947
○ = NORTH SEEKING POLARIZATION
⊡ = EARTH'S PRESENT FIELD

dark lines on a lighter background. The spacing of the surfaces is uneven and at points in a bed only a few feet apart may range from ten per inch to only one every foot. Somewhat dried-out pieces of clay often will shatter along these surfaces into a number of rhombic plates when dropped and, where the cleavage is extensively developed, the bed is more rapidly eroded than in near-by sections where it is poor. In any vertical section, all the beds have approximately the same directions of cleavage, but these directions do not persist throughout the pit.

It is considered that this cleavage was developed by the same forces that caused the folding and faulting, and because the cleavage is not the same everywhere (see Fig. 1), it is concluded that the deforming forces did not act in a uniform direction throughout the pit.

Observed and corrected magnetizations—Eleven samples from a flat-lying clay-layer were measured in order to determine the uniformity of magnetization. The results are given in Figure 2. It is seen that the declination varies over a total of 18° and the inclination over 14°. Johnson, Murphy, and Torreson [1, 1948], using the same sampling and measuring techniques, find the scatter in undisturbed Pleistocene clays to be only about ± 2°. It was clear that an accurate measure of the effect of folding could not be obtained in this disturbed location, but the measurements were carried through to determine what indications would result with regard to stability.

The fold shown in Figure 3 was sampled in the clay layers at sites indicated by

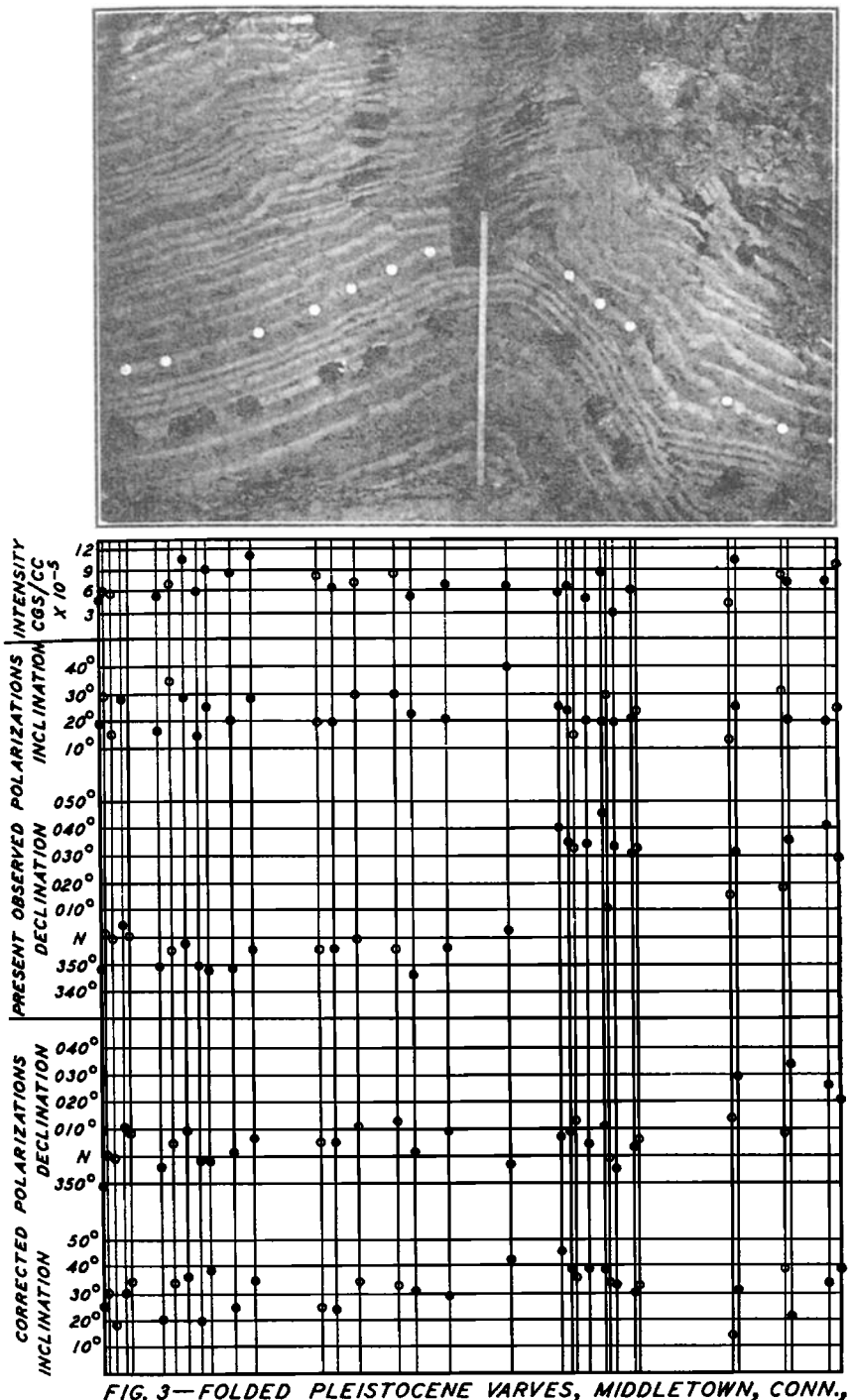


FIG. 3—FOLDED PLEISTOCENE VARVES, MIDDLETOWN, CONN., DECEMBER, 1947

sample cups and by vertical lines on the graph that have been projected downwards from the sample sites. In the upper graphs, the values of total intensity, declination, and inclination are plotted, and in the lower graphs the corrected declination and inclination are given for the case when the beds of the fold are made flat. Alternate presentation of the data is given in Figure 4 which shows, in the lower hemisphere of an equal-area projection, the relation of the observed magnetizations to the bedding poles at the sample sites, and in Figure 5 which indicates the cor-

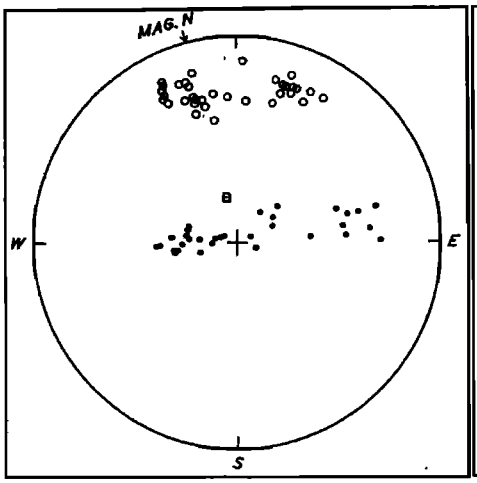


FIG. 4—OBSERVED POLARIZATIONS AND BEDDING POLES IN FOLDED PLEISTOCENE VARVES; MIDDLETOWN, CONN., DECEMBER, 1947
•= BEDDING POLES
⊟= EARTH'S PRESENT FIELD
○= NORTH SEEKING POLARIZATIONS

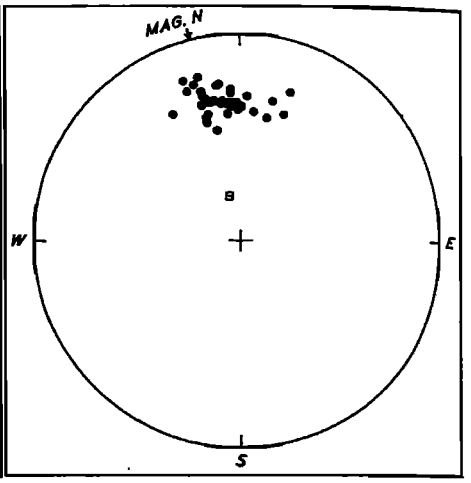


FIG. 5—POLARIZATIONS OF FOLDED PLEISTOCENE VARVES CORRECTED FOR TILT OF BEDS; MIDDLETOWN, CONN., DECEMBER, 1947
●= NORTH SEEKING POLARIZATIONS (CORRECTED)
⊟= EARTH'S PRESENT FIELD

rected values of the polarizations. The polarizations were reconstructed in the same way as was necessary to make horizontal the beds from which they were derived. This was accomplished by rotating each bed normal, first about a horizontal axis normal to the fold axis in the amount necessary to bring it into a vertical plane that is normal to the horizontal projection of the fold axis, and then rotating it about the horizontal projection of the fold axis so as to bring it into the vertical.

Discussion—Comparison of the plots of observed and corrected polarizations shows that restoring the beds to the horizontal causes the magnetizations to move towards parallelism with one another and thus, to a very rough first approximation, the polarizations seem stable. This tendency is more marked in declination than in inclination because the declination at the start of the folding is about parallel to the fold axis and the polarization is inclined below it. As the beds are rotated, the polarizations are deflected, causing the declination to change particularly and the inclination only slightly. It is felt that parallelism of the corrected magnetizations was not completely realized for the following reasons:

(*a*) The bed was not uniformly magnetized initially

(*b*) The folding did not make a uniform arch of the bed, as indicated by the

scattered positions of the bedding poles which could be determined with accuracy

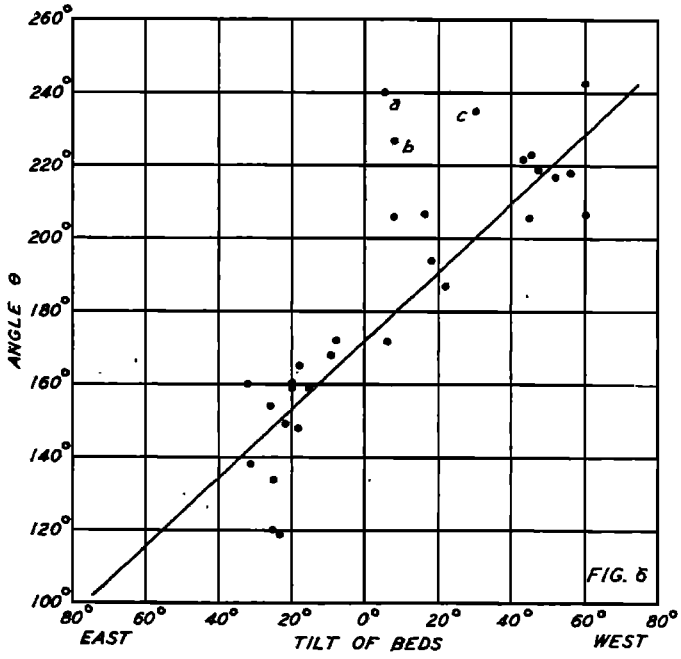
(c) The procedure of making the bed horizontal at each sample point is an approximation that applies only to the whole group

(d) Some of the beds were rotated on axes other than the fold axis, as exemplified by the samples from the lower bed of the thrust-fault zone in the right side of the exposure

(e) There are some relatively small errors in sampling, measuring, and plotting

In addition, the polarizations may have been influenced by distortion of the beds as well as by simple tilting.

A graph (Fig. 6) was constructed to investigate whether shear of the beds during folding influenced the polarizations. In this plot, θ is the angle between the north-seeking ends after the magnetizations were projected onto a plane normal to the fold axis and a reference line perpendicular to the fold axis in this plane. The various values of θ are plotted against the dips of the beds from which the samples were taken. This plot should result in a straight line having a slope of 45° if the



STABILITY DEMONSTRATED BY FOLDED VARVES, MIDDLETOWN, CONN., PLEISTOCENE VARVES, AUGUST, 1947

polarizations have not changed since the time of folding and if they were not influenced by plastic deformation. The curve would have zero slope if the polarizations are completely unstable, and a slope less than 45° if they are stable and deformation from shearing of the beds deflected them.

The straight line of Figure 6 was drawn through the averaged values of θ versus tilt of beds for the east and west dipping beds, omitting from the computation the points marked *a*, *b*, and *c*, which were from the lower bed of the thrust-fault zone in the right side of the exposure. The scatter of the points from the straight line could result from the irregularities just mentioned. The slope of the curve, 43°, indicates that, on the average, deflection of the magnetizations by shearing associated with the folding is insignificant and the polarizations are stable. No importance may be attached to the departure from 45° owing to the scatter of the relatively few points.

Conclusion—It is concluded that, in spite of the irregularities of the fold, a significant degree of stability in the polarizations of these Pleistocene varves has been demonstrated. The demonstration was not sufficiently sensitive to reveal slight changes which the polarizations may have undergone, but the evidence indicates that these clays retain substantially unchanged to the present time the magnetizations they acquired at deposition.

This conclusion differs from the opinion reached by Ising [12, p. 18, 1943] that the remanent magnetizations of flat-lying Quaternary varves in Sweden follow faithfully the earth's magnetic field. His data for declination agree well with this concept, but the observed inclinations are 20° to 35° less than the earth's present field.

Johnson, Murphy, and Torreson [1, 1948], in studies made in a disturbed clay-layer subsequent to this, found evidence that the New England clays do not follow the earth's magnetic field.

Ferromagnetic component—Some of the magnetic fraction was separated from 11.5 kilograms of the winter portion of the varves in which the stability measurements were made. A suspension of the clay was made in three gallons of water and over a period of several hours it was gently stirred, while an 8-inch saturated Alnico V bar-magnet covered with a very thin-walled rubber membrane was immersed in it. The magnetic fraction was frequently removed from the magnet and, even after several hours in the suspension, some magnetic material continued to be removed. The magnetic material was washed several times by suspension in a small volume of water and collection on the magnet. About 100 milligrams of magnetic material were obtained. It is concluded from measurements of the intensity of magnetically saturated samples of the purified and unpurified clay that 12 per cent of the magnetic fraction was removed and that the magnetic fraction thus represents 0.3 per cent by weight of the clay.

An X-ray diffraction-pattern of the purified magnetic material was obtained with a North American Phillips Geiger counter X-ray spectrometer using iron radiation. Five maxima were obtained, which agree in spacing and intensity with those from a standard sample of Mineville magnetite. No other material was detected in the diffraction pattern, thus suggesting a purity of at least 95 per cent magnetite. Broadening of the maxima indicates that a considerable proportion of the material consists of particles less than one micron in diameter.

Samples for electron-microscope observation were prepared by placing single drops of a suspension of the magnetite in 500 cc of water on standard Formvar mounts. Some samples were dried solely in the earth's magnetic field and others were dried while in a field of 50 gauss oriented parallel to the plane of the mount.

The dry samples were given shadowing with evaporated chromium, with a shadow-length to particle-height ratio of 5 to 1.

Figure 7 shows the typical appearance of the specimens at a magnification of 10,000 diameters. The magnetite particles range in diameter from 5μ down to

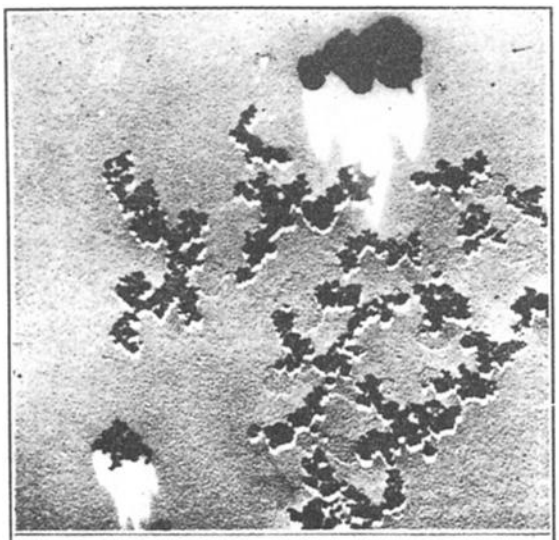


FIG. 7—MAGNETITE FROM PLEISTOCENE VARVED CLAYS; MAGNIFICATION 10,000 DIAMETERS

0.01μ, with those measuring 0.5μ to 0.1μ the most common. The particles are generally equidimensional and have irregular surfaces rather than cleavage or crystal faces. Specimens dried in the earth's field and in the field of 50 gauss appear the same. Particles of magnetite incorporated in a matrix of another mineral are not observed.

There is only a limited tendency for the particles to clump together to form extended aggregates, and these are composed mostly of the particles of the order of a micron in diameter. Particles 0.1μ in diameter and smaller are usually found as isolated individuals.

A feature in the photographs that is difficult to explain is the presence of a disperse phase that appears to show no crystalline characteristics. It is not found in all the specimens and is therefore regarded as a contamination introduced during sample preparation.

Miocene conglomerate

General statement—One mile west of the town of Selah, Washington, there is an excellent exposure (Fig. 8) of the Ellensburg formation of Miocene age, described by Smith [13, p. 3, 1903]. This deposit consists of generally undisturbed, moderately consolidated, river gravels and sands overlain and interbedded with many feet of consolidated fine-grained lake-deposits.

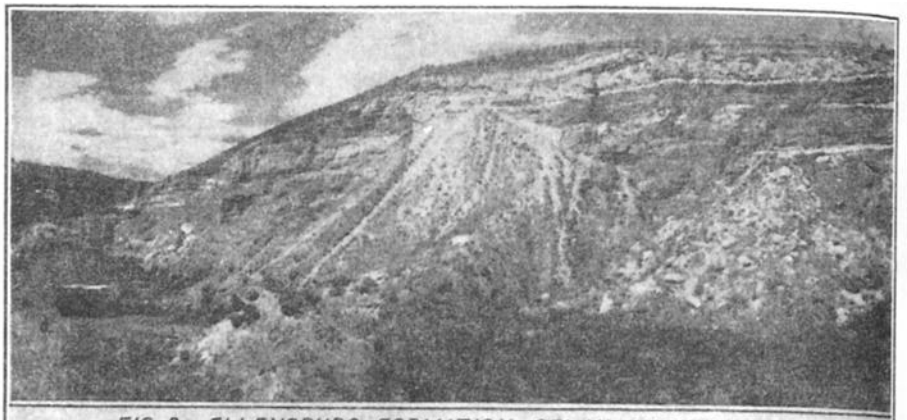


FIG. 8—ELLENSBURG FORMATION, SELAH, WASHINGTON

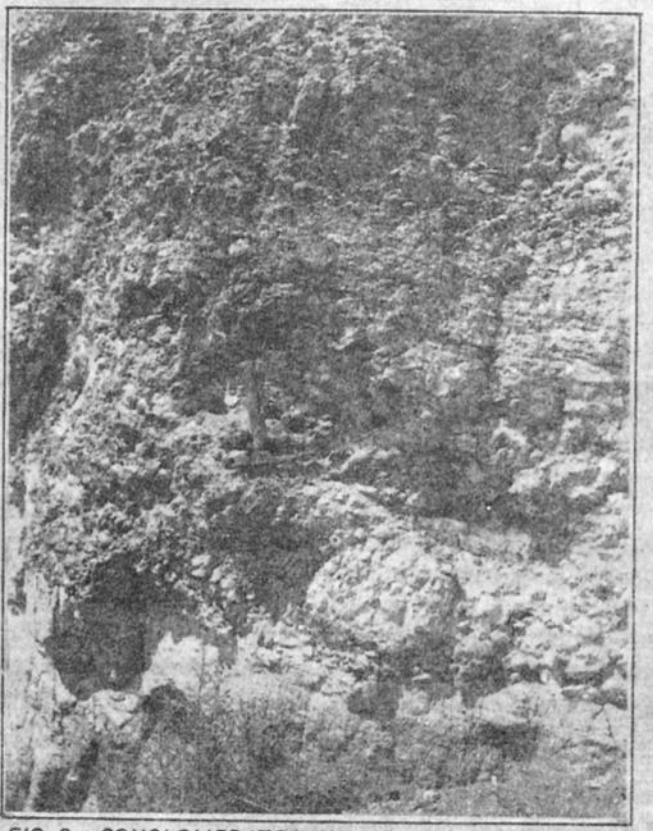


FIG. 9—CONGLOMERATE AND LAKE BEDS IN ELLENSBURG FORMATION; SELAH, WASHINGTON

The conglomerate is made up mainly of well-rounded pebbles of igneous extrusive rocks, but contains in a few patches a good percentage of angular slabs of fine-grained sediments that are lithologically similar to the lake deposits seen *in situ* in the exposure (Fig. 9). Presumably these slabs were derived from lake beds only a short distance from their present position.

Magnetizations observed—The intensities and directions of magnetization of eight of these slabs were determined by measurements on 14 cubes cut from them. The intensities ranged from 1×10^{-6} cgs/cc to 100×10^{-6} cgs/cc and averaged about 50×10^{-6} cgs/cc. The polarizations are plotted in Figure 10 in the lower hemisphere of the equal-area projection, where the numbers adjacent to the plotted polarizations correspond to sample numbers of the blocks.

Discussion—The plot shows that the various conglomerate slabs of the original lake-bed, at most a few inches in each dimension, are polarized in directions differing markedly from the earth's present field and from one another. The plot shows also that slabs 9, 11, and 13 are approximately uniformly polarized. This uniformity indicates that the slabs have not been magnetized by being in contact in the conglomerate with strongly magnetized pebbles of igneous rock.

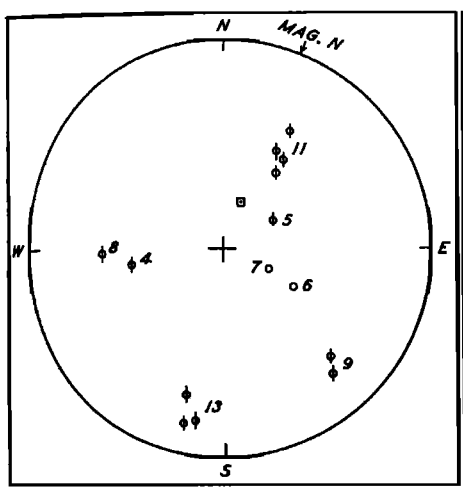


FIG. 10—POLARIZATIONS OF MIOCENE CONGLOMERATE SLABS; SELAH, WASHINGTON, JULY 21, 1948
o=NORTH SEEKING POLARIZATIONS
ϕ=SOUTH SEEKING POLARIZATIONS
⊡=EARTH'S PRESENT FIELD

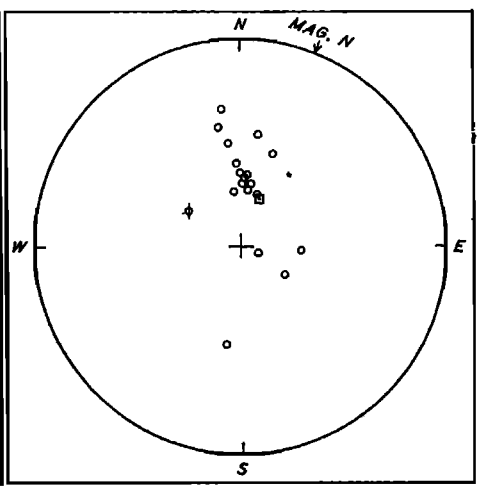


FIG. 11—POLARIZATIONS OF FLAT-LYING MIOCENE BEDS; SELAH, WASHINGTON, JULY 21, 1948
o=NORTH SEEKING POLARIZATIONS
ϕ=SOUTH SEEKING POLARIZATIONS
⊡=EARTH'S PRESENT FIELD

It is reasonable to assume that the various small slabs, after removal from their original lake-bed positions, were deposited in a generally random manner in the bed of the stream, and that the polarizations in the same way were set at different directions to one another and to the earth's magnetic field. Measurements on only eight slabs are too few to prove that the polarizations of all the slabs in the gravel are completely at random and too few to make apparent any slight tendency of the polarizations toward a preferred orientation. The measurements are sufficient, however, to show that the polarizations of these slabs change only slowly compared

with the time they have had for change since the Miocene. They would all be parallel to the earth's present magnetic field if they were completely unstable, or they would be parallel to each other if the whole deposit had been remagnetized after deposition. No mechanism has been conceived that could account for the magnetizations observed in the slabs other than the depositional process.

The polarizations of the conglomerate are in marked contrast with the polarizations of the lake beds in the same exposure which were observed by Torreson and Murphy. The author is grateful to them for permission to present their data (Fig. 11). Here half of the polarizations are almost parallel to the earth's present field, and the remainder are scattered. Small localized distortions of the beds obviously deflected many of these polarizations, but the cause of divergence of the others is not known. Stream-bedding and large particle-size are suspected.

Conclusion—It is tentatively concluded, on the basis of the few samples measured, that these Miocene sediments have retained with little change the magnetizations they acquired at the time of deposition. The data from the flat-lying beds, in addition to other data obtained on the same expedition, suggest that the earth's magnetic field in Miocene time at this locality was close to the direction of the earth's present magnetic field.

Coarse Eocene sandstone

General statement and observations—In company with Torreson and Murphy, five observations were made of the polarizations of a fluvatile deposit of gray, soft sandstone, found two miles west of Gardner, Colorado, in Huerfano Park. This deposit is regarded as of Eocene age by Osborn [14, pp. 74–75, 1929]. It is composed of non-magnetic grains up to 1 mm in diameter, and about 0.2 per cent by weight of magnetite in irregular grains which for the most part are about 0.1 mm across.

Four of the five polarizations are oriented with the south-seeking poles below the horizon and are widely scattered in declination and inclination. The fifth is 85° from the earth's present magnetic field. The magnetic moments range from 1.8×10^{-6} cgs/cc to 4.2×10^{-6} cgs/cc. Two of these samples were contiguous and the remainder came from the same bed at a distance of less than three feet.

Discussion—Dr. H. Tatel has made the suggestion that these polarizations might indicate that the conditions during deposition were so disturbing (large grains, stream-velocity gradient) that the magnetized particles, instead of achieving a preferential orientation parallel to the earth's field, settled at random. This suggestion was supported by subsequent measurements, as follows: A small quantity of the dried-out sand was moistened slightly, kneaded to make it random, and then stuffed into a sample holder for measurement. The procedure was repeated five times and the moments found range from 1.4×10^{-6} to 5.3×10^{-6} cgs/cc. These are comparable to the moments of the oriented specimens.

The same experiment was repeated with five samples of kneaded Pleistocene varved clay. Signals could not be detected, indicating that the naturally occurring clays are polarized at least 1,000 times as strongly as the same material made random.

The different results in the two experiments are explained by the great difference in the magnetic particle-size and concentration in the two sediments. The magnetic

moment of any sample is the vector sum of the moments of all its particles. Calculations show that in each 3-cm cubic sample of the clay there would be approximately 10^{10} particles of magnetite if each had a diameter of 1μ, and in the sandstone each sample would contain 10^4 particles with diameters of 0.1 mm. These two types of particles differ in volume by a factor of 10^6. Their moments probably differ by a smaller factor [4, Kittel, 1946].

It can be inferred from these values that the likelihood of finding in kneaded samples a moment greater than some small arbitrary value is much greater in the case of the coarse material having fewer magnetite particles, each of which makes a great contribution to the vector sum, than in the case of the fine material with many more much smaller particles adding their moments vectorially.

Conclusion—It is concluded that at deposition this sediment was polarized at random and not parallel to the earth's field. The polarizations were not demonstrated to be completely random by the few samples, but the scatter of the values indicates that the polarizations do not change rapidly.

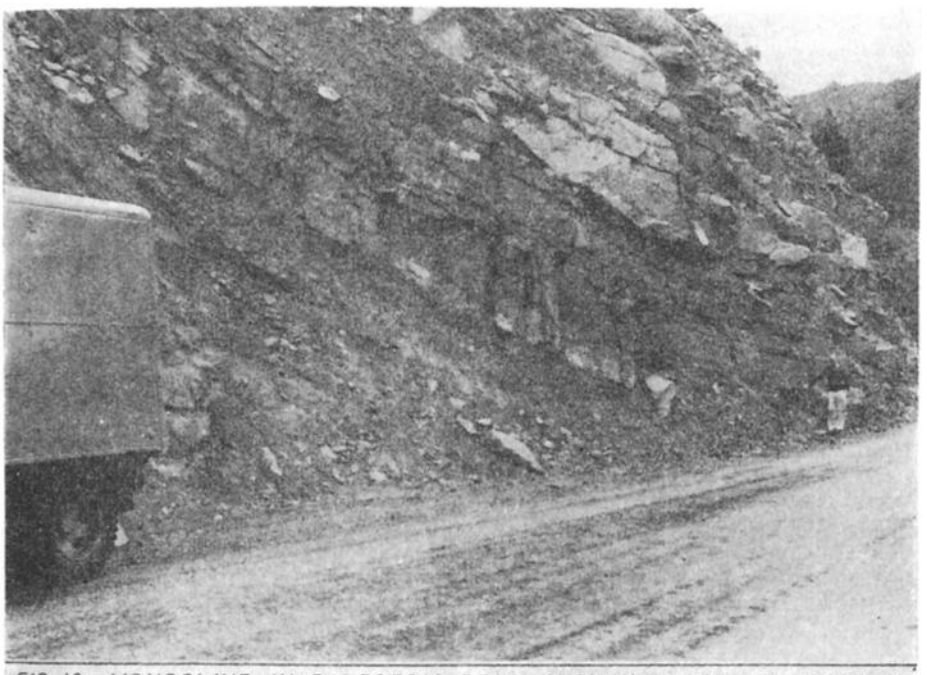


FIG. 12—MONOCLINE IN PALEOZOIC BEDS, GLENWOOD SPRINGS, COLORADO

Inconclusive study of stability

Four miles west of Glenwood Springs, Colorado, in an exposure beside U. S. Highway 6, studies were made in a monocline (Fig. 12) of late Paleozoic red sandstones, shales, and conglomerates. It was possible to remove two large blocks of fine-grained, uniformly-bedded sandstone from the same stratum, and from each block five oriented cubes were cut. Within each block the polarizations are the same

within 4°. In Figure 13, their averaged values are plotted and the poles of the bedding of each block and the average fold axis are indicated. Related polarizations and bedding poles are indicated by corresponding letters.

The polarizations are very nearly parallel to the fold axis. This, then, is an example of the degenerate case which has been pointed out as being incapable of yielding definitive results on stability one way or the other. No similar cases were encountered elsewhere in the course of this work.

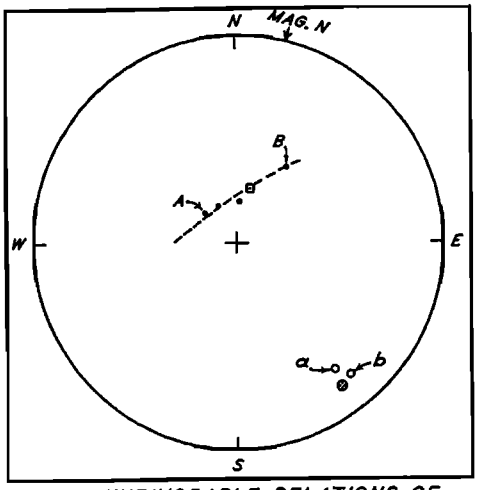


FIG. 13—UNFAVORABLE RELATIONS OF POLARIZATIONS TO FOLD AXIS; GLENWOOD SPRINGS, COLORADO, JUNE 21, 1948
○=OBSERVED NORTH-SEEKING POLARIZATIONS
⊗=FOLD AXIS
•=BEDDING POLES
⊡=EARTH'S PRESENT FIELD

Appalachian folding of ferruginous sandstone—satisfactory evidence for magnetic stability over long periods

General description—At Pinto, Maryland, nine miles south-southeast of Cumberland, the Baltimore and Ohio Railroad has made an excellent exposure of Silurian rocks. Swartz [15, p. 58, 1923] has described them in detail. A particularly good fold for the study of stability (Fig. 14) was found in the Rose Hill formation at the eastern end of the cut.

For sampling, a massive bed was selected that varies in thickness from 2 to 4 inches in a manner that has no apparent relation to the folding. This bed weathers to a reddish brown, but is dark blue-gray when fresh. Thin sections (Figs. 15 and 16) show that the rock consists of siderite, quartz, and magnetite. Other constituents are insignificant. The quartz, constituting about one-third the volume of the rock, occurs as angular to sub-angular, somewhat elongated, clear detrital grains of uniform size, generally oriented parallel to the bedding. These grains lack secondary-growth aprons and show no signs of having been deformed *in situ*. They are generally separated from one another by siderite, which makes up the bulk of the bed. The

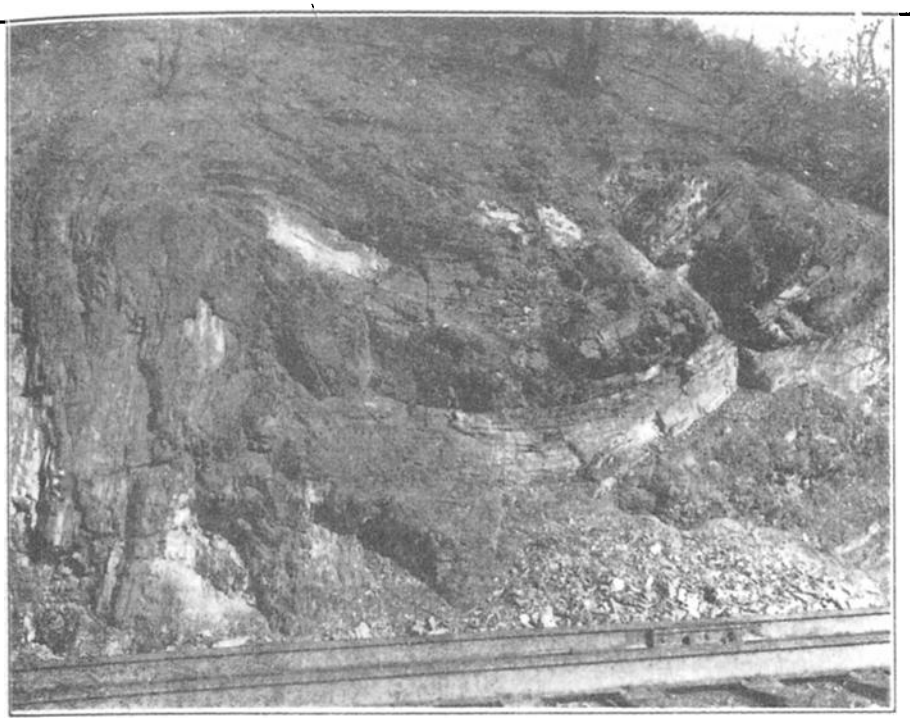


FIG. 14—FOLDED ROSE HILL FORMATION; PINTO, MARYLAND

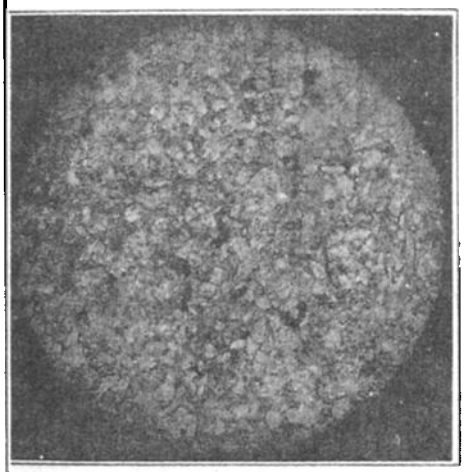


FIG. 15—FERRUGINOUS SANDSTONE, ROSE HILL FORMATION (PLANE POLARIZED LIGHT); MAGNIFICATION 46 DIAMETERS

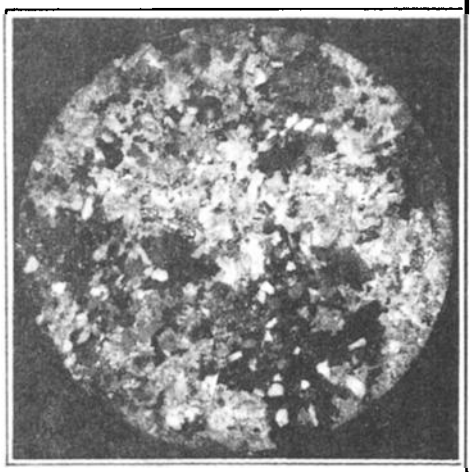


FIG. 16—FERRUGINOUS SANDSTONE, ROSE HILL FORMATION (CROSSED NICOLS); MAGNIFICATION 46 DIAMETERS

siderite has crystallized to large anhedral colorless or gray transparent patches that usually embrace several quartz-grains. Twin lamellae are rare, and no evidence of post-crystalline deformation has been noted. On weathering, the siderite grades progressively through a deeper greenish-gray to yellow into a practically opaque

deep-red material that may be a mixture of limonite and hematite. The quartz and magnetite remain as transparent and opaque islands in the weathered ground-mass. In the fresh ground-mass, with transmitted light, the magnetite is recognized as opaque, moderately well-rounded to angular, somewhat elongated detrital grains, ranging in size from the limits of resolution of the light microscope up to 0.08 mm across. With reflected light, it is recognized as bluish, glistening, pitted grains. There are some elongated aggregates of magnetite that may have coalesced during settling, but they show no alignment parallel to the projection of the magnetization on the plane of the section.

The origin of this unusual rock is not known in detail. It is significant, however, that this bed was deposited in water, as shown by its uniformity, its lack of cross bedding, its marine-fossil content in the bottom 1/2 inch (ostracods and pelecypods), and its interbedding with shales of marine origin.

The manner and time in which this bed received its magnetization are unknown. The magnetite presumably was oriented by the earth's field at deposition, but the contribution made by the siderite has not been determined. Information on the ferromagnetic properties of siderite has not been found in standard reference tables.

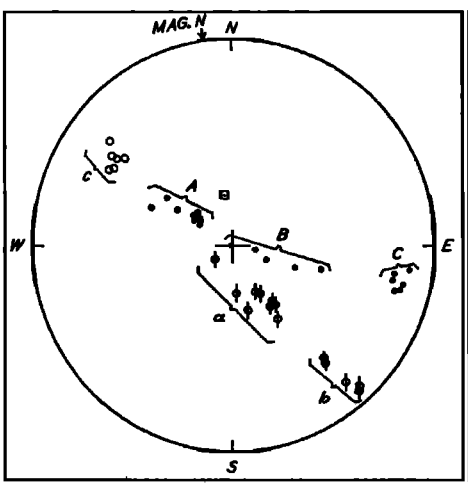


FIG. 17—OBSERVED POLARIZATIONS AND BEDDING POLES IN FOLDED ROSE HILL (SILURIAN) FORMATION; PINTO, MARYLAND, NOVEMBER 23, 1948

•=BEDDING POLES
⧫=SOUTH SEEKING POLARIZATIONS
o=NORTH SEEKING POLARIZATIONS
⧈=EARTH'S PRESENT FIELD

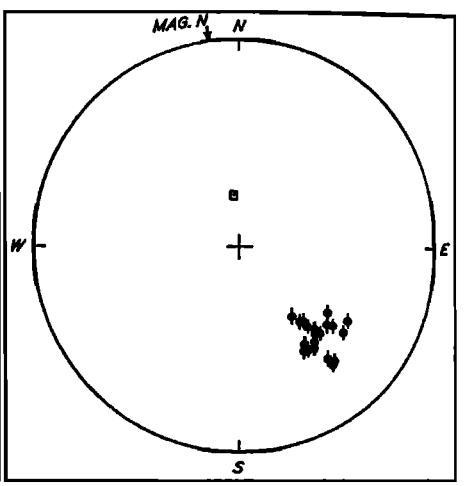


FIG. 18—POLARIZATIONS OF FOLDED ROSE HILL (SILURIAN) FORMATION CORRECTED FOR TILT OF BEDS; PINTO, MARYLAND, NOVEMBER 23, 1948

⧫=SOUTH SEEKING POLARIZATIONS (CORRECTED)
⧈=EARTH'S PRESENT FIELD

Measurements on a sample of $FeCO_3$ prepared by mixing $FeCl_2$ and Na_2CO_3 solutions suggest that siderite and its oxidation products may contribute a significant part to the moment.

These Silurian beds were deformed [15, Swartz, p. 21, 1923] near the end of the Paleozoic era. Holmes [16, p. 145, 1946] estimates the beginning of the Permian to date back about 220 million years, and the Silurian about 350 million.

The folding process appears merely to have flexed this bed gently into its present form without causing apparent shearing or flowage in the bed itself. The shales above and below show extensive distortion and contain well-developed cleavages, but this bed lacks any trace of cleavage. Slickensides crossing the bed were noted at only one place, whereas they are abundant in the shales. Three sets of joints transect the bed and the shales alike. The bed appears to have behaved as a generally coherent and passive plate that was floated about by the movements of the many feet of enclosing shale as they were squeezed into their present configuration.

Observed magnetizations—Figure 17 is a plot of the magnetizations in the fold and the poles of bedding at the sample sites. The related groups of bedding poles and polarizations are indicated by corresponding large and small letters. Points within related groups do not necessarily correspond in position in the two groups. The greatest angle between south-seeking poles is 127°, and the greatest angle between bedding poles is 113°.

Corrected magnetizations—In Figure 17, the disposition of the bedding poles shows that all parts of the bed are not parallel to a single-fold axis; instead, on the average, they seem to be wrapped into a cone. For this reason, errors, estimated to be smaller than ± 3°, arise from uncertainty of the proper way to carry out rotations to make the bed flat. The corrected magnetizations in Figure 18 were obtained by rotating the observed polarizations in the same manner as was necessary to move their corresponding bedding poles directly to the center of the projection. The appearance of the cluster of polarizations obtained by this procedure is very similar to the results obtained when a number of different combinations of fold axes were tested.

Figure 18 shows that the total scatter of the corrected polarizations from a common direction is less than ± 12° and indicates the stability of the magnetization. Each point in the group was subject to the following experimental errors whose influence on the direction of the corrected polarization have been estimated: Measuring the strike and dip of the bed, ± 3°; inscribing orientation marks on the sample, ± 1°; drilling a core perpendicular to the strike and dip lines and cutting a cube, ± 2°; determining the relation of the magnetization to the cube edges, ± 1°; plotting the magnetic data and rotating them to obtain the observed magnetization, ± 1°; rotating the observed magnetization to obtain the corrected magnetization, ± 3°. These figures give an estimated probable error for each point of ± 5°.

Conclusion—The close relation between the bedding orientation and the directions of magnetization is presented as satisfactory evidence that this rock has retained its direction of magnetization since a time prior to its folding 200 million years ago.

Regional consistency of Rose Hill formation

To investigate the regional consistency of the Rose Hill formation, samplings were made 32 miles east of Pinto at localities *A-E* (Fig. 19) along the road between Berkeley Springs and Great Cacapon, West Virginia. The geology of this region has been described in detail by Stose and Swartz [17, 1912] and Swartz [15, p. 85, 1923].

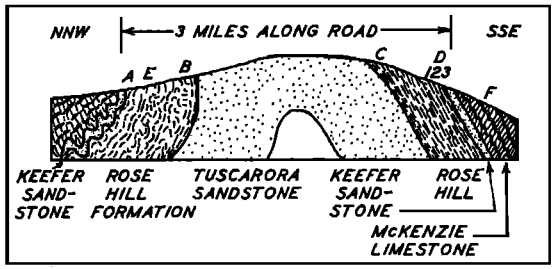


FIG. 19—DIAGRAMMATIC CROSS SECTION THROUGH CACAPON MT., WEST VA.

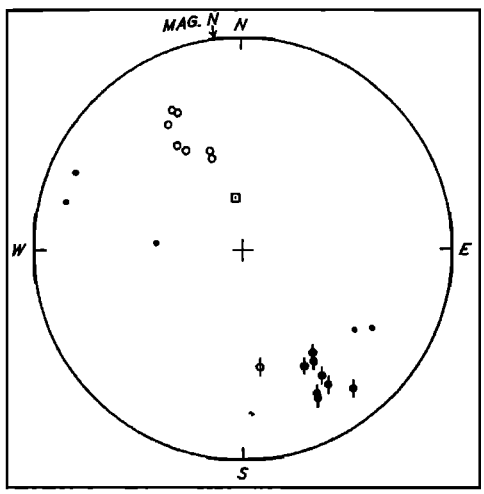


FIG. 20—OBSERVED POLARIZATIONS, CORRECTED POLARIZATIONS, AND BEDDING POLES IN ROSE HILL (SILURIAN) FORMATION; CACAPON MT., WEST VA., OCTOBER 17, 1948

○=OBSERVED NORTH-SEEKING POLARIZATIONS
⊕=OBSERVED SOUTH-SEEKING POLARIZATIONS
⊕=SOUTH SEEKING POLARIZATIONS (CORRECTED)
••=BEDDING POLES
⊡=EARTH'S PRESENT FIELD

Figure 20 indicates the results of observations in a small fold at location *E*. Samples were so difficult to remove that few measurements could be made, but the limited results indicate stability and the corrected polarizations are consistent with those found at Pinto.

Samples of the Rose Hill formation were taken at localities *A-D* (Fig. 19) in the hope of obtaining more data on the regional consistency. This procedure was considered reasonable on the basis of a recent study of Tertiary rocks by Torreson, Murphy, and Graham [11, 1949], which indicated that, in flat beds, geographic and stratigraphic positions have little bearing on the polarizations observed. The results of the samplings, limited by the inadequacies of the exposures, are given in Figure 21. Sampling and polarization-correcting procedures were the same as those used at Pinto, and the corrected polarizations are indicated in Figure 22. Letters beside

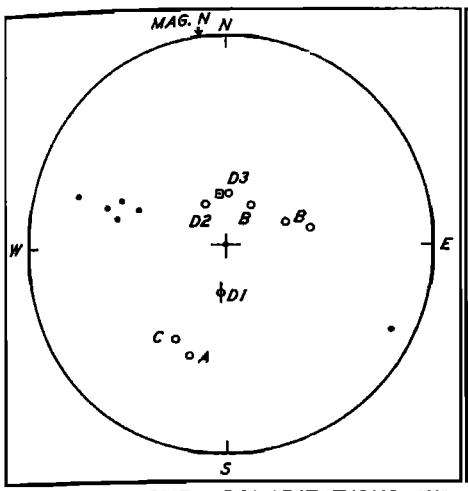


FIG. 21—OBSERVED POLARIZATIONS AND BEDDING POLES IN ROSE HILL (SILURIAN) FORMATION; CACAPON MT., WEST VA., OCTOBER 30, 1948

o=OBSERVED NORTH-SEEKING POLARIZATIONS
•=OBSERVED SOUTH-SEEKING POLARIZATIONS
••=BEDDING POLES
⊡=EARTH'S PRESENT FIELD

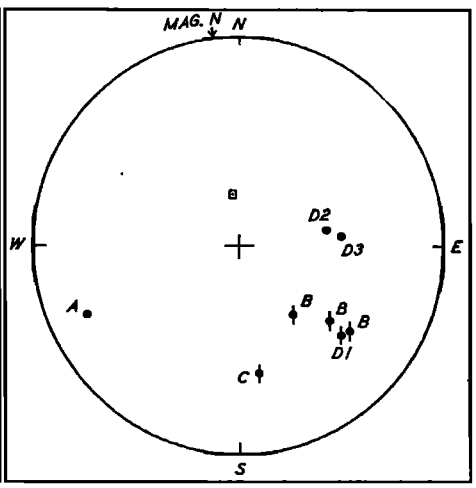


FIG. 22—POLARIZATIONS OF FOLDED ROSE HILL FORMATION CORRECTED FOR TILT OF BEDS; CACAPON MT., WEST VA., OCTOBER 30, 1948

•=NORTH SEEKING POLARIZATIONS (CORRECTED)
•=SOUTH SEEKING POLARIZATIONS (CORRECTED)
⊡=EARTH'S PRESENT FIELD

plotted points designate sample localities. *A* is about 6 feet, and *D*1, *D*2, and *D*3 about 32, 27, and 25 feet below what is regarded as the conformable contact of the Keefer with the Rose Hill. *B* and *C* are both about 15 feet above the top of the Tuscarora sandstone.

The samplings at *B*, *C*, and *D*1 are consistent with the findings at Pinto and at *E*, but the observations at *D*2, *D*3, and at *A* are anomalous, in that the north-seeking pole is now below the horizon and the declination is much changed. The polarizations observed at *F* in the overlying McKenzie formation are also oriented with the north-seeking poles below the horizon.

In spite of these inconsistencies, a tendency towards uniformity of polarizations in the bulk of the Rose Hill formation over long distances is clearly indicated. It might be remarked that, although this reversal of the polarizations is disconcerting, it may not be an unusual feature of old rocks, particularly for limited periods of time. My physicist colleagues at the Department of Terrestrial Magnetism expect to find evidences, if the history of the earth's magnetic field can ever be well established, for stronger secular-variation foci (disturbances of the general magnetic field of the earth over local regions) in remote epochs when the interior of the earth may have been hotter and the disturbing current-systems nearer the surface. These foci at present are relatively localized and move slowly over the surface of the earth; the corresponding current-systems now appear to be roughly at 1,200 km depth.

Earth's magnetic field in Silurian time—Although it is not within the scope of the present study to investigate past directions of the earth's magnetic field, it is

nevertheless of interest to consider whether the reconstructed direction of magnetization can be regarded as an indication of the direction of the field at the time of deposition of the bed 150 million years before the folding. If the polarizations have remained stable for the past 200 million years, they may be considered likely to have done so for the 150 million years preceding their deformation. If later measurements show uniformity of polarization of the major bulk of the Rose Hill formation over long distances, the argument for continued stability would be considerably strengthened and the magnetization would be shown to be something other than a local anomaly caused by unknown forces.

The approximate agreement of the reconstructed compass-direction over a distance of roughly 50 miles is striking and difficult to account for on an accidental basis. With my physicist colleagues at the Department of Terrestrial Magnetism, various efforts to reconstruct electrical current systems in or near these folds to account for the observations have led to what they regard as completely implausible hypothetical conditions. Agreement has been reached in our discussions that even the present evidence, which will be extended by further studies already under way, is exceedingly difficult to understand on any basis other than the assumption that the beds were magnetized either at the time of deposition or by the growth of magnetizable minerals in a regionally uniform magnetic field prior to the folding. No mechanism is known to the geophysicist by which a uniform magnetic field of this kind over a distance as great as 50 miles can be produced by electrical current systems (for example, associated with volcanic action). The reconstructed residual-polarization is, therefore, accepted as a measure of the direction of the earth's magnetic field in this locality at some time prior to the Permian, and very probably at the time of deposition.

Appalachian folding of limestone

General statement—To investigate the stability relations in a deformed rock of different lithology, observations were made at *F* (Fig. 19) in the closely-folded McKenzie formation about 200 yards northeast of *D*.

The formation here consists predominantly of buff weathering and blue, dense, fine-grained, fossiliferous, thin-bedded limestone, with a small proportion of interbedded shale. Figure 23 shows the detail of a typical fold in the exposure. The straight parts of the limestone appear substantially undeformed. Flow and fracture cleavage are absent in the limestone and only poorly developed in the shale; joint planes are spaced inches apart and are filled with secondary calcite; slickensides are common only in the shales; the numerous fossil-shells in the limestone appear undeformed in the hand specimen, and under the microscope are seen to consist of a "comb structure" (Fig. 24) of relatively large, unstrained and only moderate-twinned, anhedral calcite-crystals, whose *c*-axes are normal to the surface of the shell. The ground-mass of these shells is a disorganized aggregate of calcite crystals less than 0.01 mm in length with other fine-grained, undetermined components. Magnetite occurs as small, irregular, detrital grains or coalesced, elongate shreds up to 0.05 mm long.

Observations—Figure 25 shows the observed and corrected polarizations and the bedding poles of the sample blocks. Sampling and polarization-correcting

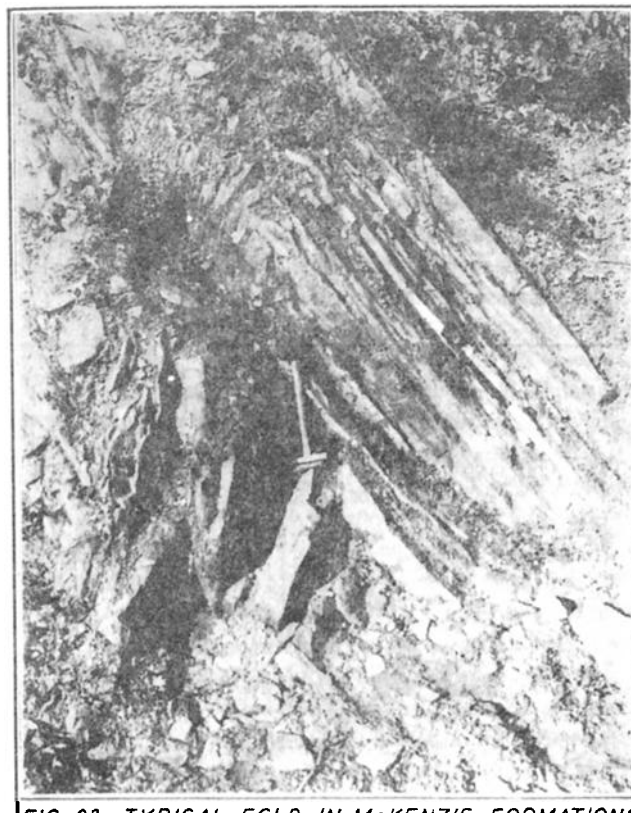


FIG. 23--TYPICAL FOLD IN McKENZIE FORMATION; CACAPON MT., WEST VIRGINIA



FIG. 24—THIN SECTION OF LIMESTONE IN McKENZIE FORMATION (CROSSED NICOLS); MAGNIFICATION 46 DIAMETERS

procedures were the same as those used at Pinto. Accurately oriented samples could not be obtained from the crest of the fold. The observed polarizations form a group with a maximum scatter of ± 35°. This group is divided into two distinct groups when the beds are made flat, the arrows indicating the directions the polarizations move when the restoration is made.

Discussion—A specific kind of "instability" of the polarizations is suggested by their pattern in the deformed bed. By folding, the bedding poles were deflected a maximum of 165° and the maximum angle between polarizations is only 70°. This latter spread is not in the direction that the folding would be expected to produce. Instead, it is in planes parallel to the fold axis.

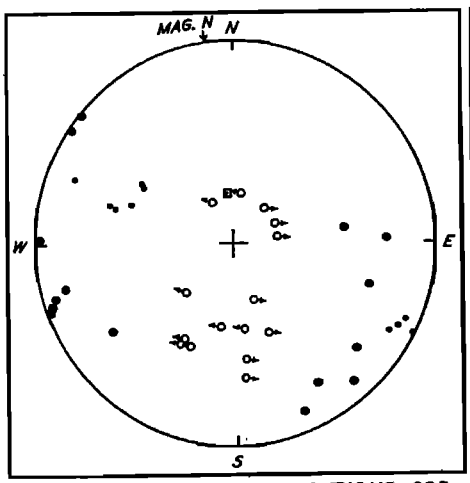


FIG. 25—OBSERVED POLARIZATIONS, CORRECTED POLARIZATIONS, AND BEDDING POLES IN McKENZIE (SILURIAN) LIMESTONE, CACAPON MT., WEST VA., OCTOBER 21, 1948

o=OBSERVED NORTH-SEEKING POLARIZATIONS
•=NORTH SEEKING POLARIZATIONS (CORRECTED)
••=BEDDING POLES
■=EARTH'S PRESENT FIELD

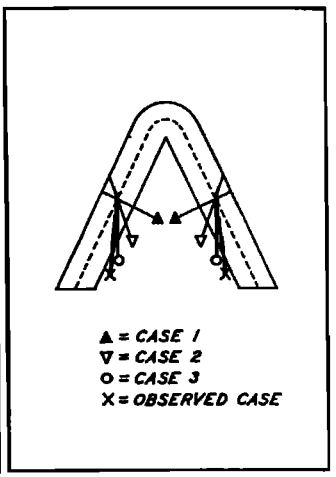


FIG. 26—EFFECT OF PLASTIC DEFORMATION ON POLARIZATIONS

It can be imagined, no doubt, that immediately after the folding, the polarizations were in a completely different pattern from now, and that at some time during the past 200 million years they moved towards parallelism with one another in a field that differed perhaps from the earth's present magnetic field. Although such may have been the case, the scatter of the polarizations shows that they are not completely unstable.

In the folds studied previously, no evidence of deflection of the polarizations by shearing of the beds during folding was found, and it has been demonstrated that the polarizations of the Rose Hill formation have remained stable for at least 200 million years. It is interesting to consider whether the pattern of polarizations in the McKenzie formation might be attributed, not to instability, but instead, to the deformation, particularly as a result of shear in these lime-beds.

Figure 26 is a schematic construction of the positions that the projections of

the polarizations on a plane normal to the fold axis would acquire if the bed yielded to deformation in various ways. The polarizations are assumed to be deflected by plastic deformation in the same way that any reference direction is influenced in a body that is deformed; see Nadai [18, pp. 291–307, 1931]. In this hypothetical picture, which is based in part on the features in the fold, these projected polarizations are normal to the bed at the beginning of the folding. Case 1 is for polarizations in a bed that folds without shearing. For each degree of tilt of the beds, there is a corresponding degree of tilt in the projected polarization (evidently the case in the varved clays, and more clearly in the ferruginous sandstone). In Case 2, the polarizations are deflected by simple shear that arises in the limbs of the fold as a result of the folding. The bed maintains constant thickness, and the situation is analogous to a pad of paper being bent. In Case 3, the bed acts like two sets of parallel rulers, attached end to end by hinges, the points of attachment representing the crest of the fold. As the bed is hinged about the crest, the bed thickness decreases by combined shear, compression, and outward flow, but the projected polarizations, like the braces of the rulers, remain parallel no matter to what extent the limbs are bent towards each other. For comparison with these cases, the projections of the observed polarizations are shown schematically.

Case 3 approaches most closely the observed case. For Case 3 to take place, a considerable increase in area of the bed would be necessary to maintain the volume and compensate for the decrease in thickness. Presumably this would be accomplished mainly by extension upward along the bedding. A further decrease in bed thickness than is postulated for Case 3 would cause the lower ends of the projected polarizations to diverge from one another, in a manner shown by the observed case.

A complete extension of this argument into three dimensions becomes more complex than the data warrant. Stated briefly, however, the projections of the polarizations on the axial plane of the fold could be widely scattered by non-uniformity in the direction of upward extension of the bed. This non-uniformity would not cause extensive changes in the polarizations projected on the plane normal to the fold axis.

The foregoing hypothesis, based on geometric relations and the assumptions that the polarizations are stable and are deflected in accordance with predictions of the theory of plastic deformation, could explain how deformation might be responsible for the observed pattern of polarizations in some beds, while other beds, subjected to a quite different physical pattern of deformation, would behave as observed in the Rose Hill formation. Intermediate cases should exist.

Geological evidence supporting the postulated behavior of these particular lime-beds is lacking. No such drastic changes in bed thickness or extension of the beds took place in these rocks when they were in a solidified condition, as shown by the undeformed fossils and the lack of cleavage. In order that deformation could account for the pattern of polarizations, it would be necessary that at the time of folding the sediment be in a relatively "soft" condition. Cloos [19, p. 876, 1947] has found evidence that certain other sediments in the Appalachians through part, at least, of their folding, were not consolidated.

Further work is necessary to determine the stability of these polarizations and the influence of the deformation on them.

Conclusion—From the behavior of the polarizations of the limbs of the fold when the beds are made flat, it is concluded that the polarizations observed for these lime-beds do not give a simple indication of the magnetization of the flat bed prior to its folding. It has not been determined whether the pattern of polarizations resulted from the folding process or in some other way.

IV—SUMMARY STATEMENTS

This study was undertaken to investigate whether sedimentary rocks can retain without change their directions of remanent magnetization through long periods of geologic time.

The problem of determining the stability of the magnetism in rocks during very long periods of time was formulated in a way which is accessible to measurement. Experimental procedures and suitable measuring and sampling techniques were developed, in collaboration with others.

Investigations of stability were carried out in field work extending over large areas and on rocks of ages extending back to the Silurian. A discovery was made of new evidence which is considered satisfactory proof that the magnetizations in certain rocks have remained permanent over periods ranging from 15,000 years to over 200,000,000 years. Conflicting evidence and views on this point have appeared in the literature over the past 50 years, but these have been based on inadequate technical methods.

Indications have been found of a regional similarity in the magnetization of a Silurian formation over a moderate distance (of the order of 50 miles before the folding). This uniformity cannot be reasonably accounted for by remagnetization processes arising from electrical current systems or by any process except the growth of magnetic minerals in a regionally uniform magnetic field (earth's magnetic field) prior to the folding, or by the original deposition of magnetic particles in the Earth's magnetic field.

One method of determination has been to observe the magnetizations in a bed that long ago was deformed in such a way that the polarizations at various points in the bed became oriented in different directions relative to one another and to the direction of the earth's field. If the polarizations in such a bed are magnetically unstable, they will shift toward parallelism with one another throughout the deformed bed. If the polarizations are stable, and are not shifted by plastic-flow processes, they will maintain a constant relationship to the bedding, during and after deformation, such that when the bed is reconstructed to its original flat condition the polarizations will once again become parallel.

An alternate procedure has been to look for random polarizations in beds that were laid down in a random manner. The premise is that, if in a conglomerate, for example, all the pebbles have the same directions of magnetization, then these polarizations must be regarded as ones that were acquired after the pebbles became imbedded; but if the polarizations in the pebbles are dispersed, stability of the magnetization of each pebble is indicated.

Studies in Pleistocene clays that were folded shortly after their deposition show, within limits of error which appear reasonable for disturbed beds, that for each degree the beds were tilted in the deformation, the polarizations on the average were tilted correspondingly. This fact indicates that the polarizations have remained

largely unchanged, and that in the deformation, shearing of the beds was not appreciable. On restoring the beds to the flat condition, the polarizations did not become perfectly parallel because of irregularities in the folding and the initial magnetization. The magnetic fraction of the clays is described.

Pebbles of a Miocene lake-deposited sediment imbedded in a conglomerate were found to have scattered polarizations. On the other hand, the polarizations in the undisturbed source-beds of these pebbles were found to cluster together on the projection plot. The conclusions are reached that the polarizations of the pebbles and the lake beds have remained stable since the time of deposition, and that the cluster of directions of magnetization represents the approximate direction of the earth's magnetic field in Miocene time. A coarse-grained Eocene sandstone was found to have scattered polarizations. Tests demonstrate that this distribution of the polarizations could be the result of disturbing conditions during deposition. The conclusion is reached that, although the polarizations have no relation to the earth's field in Eocene time, they nevertheless have remained stable.

The polarizations in a Silurian bed, folded near the end of the Paleozoic era, are found to maintain systematic angular relationships to the bedding throughout a severe fold. When the bed is reconstructed to its flat condition, the polarizations all become approximately parallel, indicating that they have not changed since the folding 200 million years ago. This behavior of the polarizations and field evidence show that shearing in these beds during folding was slight.

At a distance of 32 miles from the exposure where stability in the Silurian beds was demonstrated folds in the same formation give comparable results, except in the uppermost layers.

Studies in a closely-folded Silurian limestone were inconclusive. The pattern of polarizations does not conform to the folding, and the cause has not been determined. The possibilities are considered that either true magnetic instability or shearing and extruding processes during deformation were responsible for the observed pattern.

The chief contribution here presented is evidence of a new kind for accepting the conclusion that the direction of the earth's magnetic field, as it existed in remote geologic time, is preserved in certain rocks which can be identified as satisfactory for this purpose.

A thorough study of many examples of such rocks has not been possible as yet, and will require several years of field studies. Other examples of appropriate rocks must be located and tested by finding suitable folds, and regional uniformities must be determined. The original goal of this study was to determine whether any sedimentary rocks can be definitely shown to preserve their directions of magnetization over very long periods of time and during deformation.

This demonstration may provide a basis for geological studies of folding and deformation processes, as well as other new evidence bearing on the history of the earth.

V—FINAL CONCLUSIONS

Sedimentary rocks have been found which have retained a direction of magnetization through long periods of geologic time, and it has been shown possible to identify rocks which have this characteristic. The field studies which have been made

indicate that a useful degree of magnetic stability is not an unusual property of sedimentary rocks.

VI—FUTURE WORK

This newly recognized property of rocks suggests various important investigations that may be listed in the form of questions:

(1) Can the general rule be formulated that all sedimentary rocks which have not been metamorphosed retain their primary magnetizations?

(2) Will it be possible to trace back in geologic time the history of the earth's magnetic field?

(3) Will the study of rock magnetism be useful in throwing light on questions regarding large-scale movements of the crust, such as continental drift and polar migration?

(4) Will rock magnetism be useful in describing the distortions rocks have undergone, and thus help in the understanding of tectonic features such as cleavage, lineation, and jointing?

(5) Will rock magnetism have value in the interpretation of complex structures of the crust?

(6) Will rock magnetism be useful in stratigraphic dating?

Some of these questions are already under investigation at the Department of Terrestrial Magnetism. The writer has been invited to join in these studies under a special grant from the Carnegie Institution of Washington.

VII—ACKNOWLEDGMENTS

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Appendix A

GRAPHIC SOLUTION OF SOLID GEOMETRY IN STUDIES OF ROCK MAGNETISM

The study of the directions of magnetization of rocks involves numerous problems in solid geometry, and in all situations so far encountered a graphic method of solution of these problems has proved to be rapid and sufficiently accurate. Of the many graphic methods of showing on a plane the relationships in space of lines and planes, the equal-area projection has been chosen as most suited to the requirements of the work. This projection is similar in many respects to the better-known stereographic projection. There are many discussions of the equal-area projection and its applications to various problems[1,2,3], but none is completely suited to the requirements of the present work. In the following paragraphs, therefore, the most frequently encountered problems in studies of rock magnetism are solved and the possibilities of the equal-area projection illustrated.

With the equal-area projection, it is possible to show graphically on a plane the *angular* relationships in space of planes and lines. The basis on which the projection

[1]W. Schmidt, Gefügestatistik, Tschern. Mitt., 38, 393-423 (1925).

[2]E. B. Knopf and E. Ingerson, Structural petrology, Geol. Soc. Amer., Mem. 6, 270 pp. (1938).

[3]H. W. Fairbairn, Petrofabrics, 143 pp., Cambridge (1942).

is carried out is shown in Figure 27. The direction OP passes through the center of the sphere of projection and pierces the sphere at point P. The plane of projection is tangent to the sphere at point B. The equal-area projection P' of the direction OP is the point at which the arc of radius BP, drawn from B as a center in the plane BOP, pierces the plane of projection. In this manner, all directions in space are plotted as points on the plane of projection. The equatorial plane of the sphere of projection is projected as a circle (the circle of projection) and any direction parallel with this plane projects as a point on its periphery. A line normal to the plane of projection plots as a point in the center of the circle.

A plane in space is plotted in much the same manner; a plane parallel to it is passed through the center of the sphere of projection, and then its intersection with

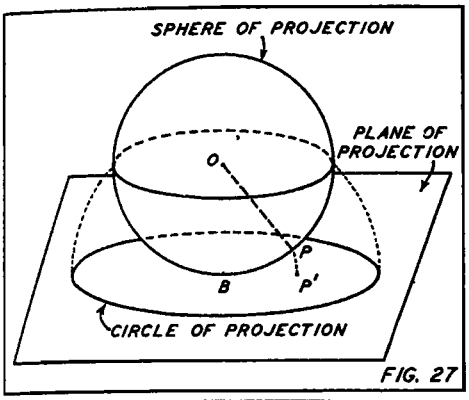


FIG. 27

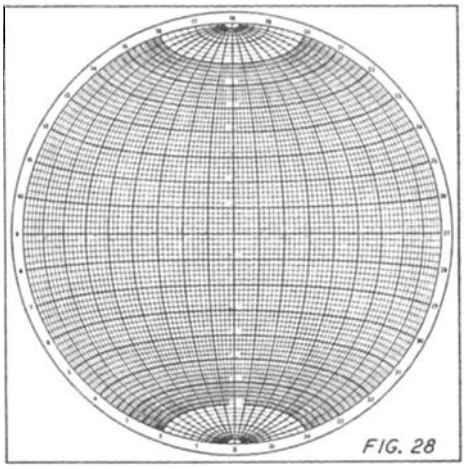


FIG. 28

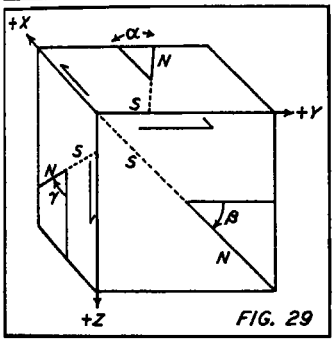


FIG. 29

FIG. 27—CONSTRUCTION OF EQUAL-AREA PROJECTION P' OF DIRECTION OP

FIG. 28—EQUAL-AREA NET

FIG. 29—COORDINATE AXES AND MAGNETIZATION VECTOR PROJECTED ON THREE CUBE FACES

the sphere is projected onto the projection plane. It is known that the projections of planes in the equal-area projection are all arcs of circles of a radius equal to or greater than that of the projection circle, with the two ends of the arc cutting the projection circle at diametrically opposite points.

The projection of direction OP in Figure 27 has been carried out by using the pole B of the sphere of projection and restricting the direction OP to the lower hemisphere of the sphere of projection. This operation is known as plotting in the lower hemisphere. By extending the direction OP to the upper hemisphere and projecting it on a plane tangent to the upper hemisphere, however, a different plot will be obtained than when the lower hemisphere is used for projection. Therefore,

it is mandatory during the solution of a problem to use consistently one hemisphere. In studies of rock magnetism, the lower hemisphere has been used exclusively, in accordance with the custom in magnetic work of thinking of the earth's magnetic field as dipping into the globe.

Figure 28 illustrates an equal-area net for use as a graphic aid in plotting. It represents the circle of projection. Over this grid a sheet of tracing-paper is placed and allowed to rotate by a pin through the center or by some more elaborate mount rotating concentrically with the projection circle. This net has been chosen in preference to the Wulff stereographic net commonly used by crystallographers for the reason that a unit area anywhere within the net embraces an almost constant solid angle. This feature is valuable in the statistical study of the distribution of directions.

The family of arcs of circles on the net represents the projections of a bundle of planes spaced at two-degree intervals, all passing through the same diameter of the projection sphere. These projections are known as great circles. The other family of curved lines consists of "small circles" which may be considered to represent the equal-area projection of the intersections of a number of parallel planes, perpendicular to the plane of projection, with the sphere of projection. They are used when the orientation of coordinate axes is changed.

Determination of the direction of magnetization of a cube—In order to treat the relationships of directions in space, it is necessary to establish a system of coordinates relative to which all measurements are made. In the work on rock magnetism where, in addition to having directions in space, quantities have a sense, that is, north or south, up or down, it is essential to establish a convention by which all angles are measured. In keeping with geographic practice, positive angles are measured clockwise, and the convention has been adopted to carry out the magnetic measurements relative to the north-seeking side of the sample. For the coordinate axes, a right-handed orthogonal system is used, with $+X$, $+Y$, and $+Z$ being related as are the thumb, index, and middle fingers, respectively, of the right hand. $+X$ customarily designates north, $+Y$ the east, and $+Z$ is then directed downwards.

Three mutually-perpendicular cube-edges are labeled with the coordinate axes as shown in Figure 29. Angles giving the positions of the projections of the magnetization vector on the three coordinate planes are defined as follows:

α is measured clockwise from $+X$ toward $+Y$ to the N seeking end of the projected vector

β is measured clockwise from $+Y$ toward $+Z$ to the N seeking end of the projected vector

γ is measured clockwise from $+Z$ toward $+X$ to the N seeking end of the projected vector

These angles fix the orientations of the traces of three planes, each parallel to a cube edge, which contain the north-south direction of magnetization. α, for example, defines the orientation of the plane that is parallel to both the Z axis and the direction of magnetization. The three planes defined by α, β, and γ must intersect in a direction in space that is the direction of magnetization.

To determine the magnetization of the cube, angles α, β, and γ (as though representing the traces of planes) are plotted relative to $+X$, $+Y$, and $+Z$, as in the example of Figure 30. The three planes should pass through a common line (point in projection) of intersection. Since the half planes containing the north-seeking halves of the magnetization vector were used in the measurements of the angles,

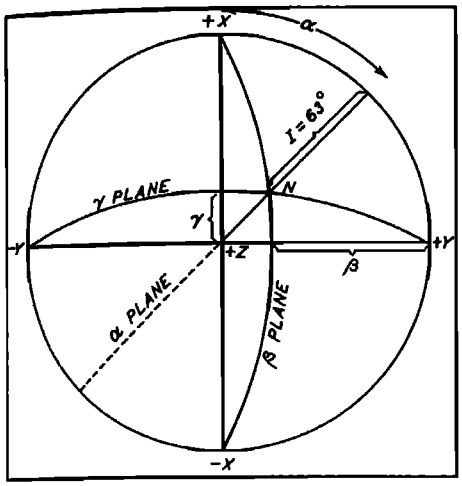


FIG. 30—PLOT OF THE DIRECTIONS OF THE MAGNETIZATION VECTOR PROJECTED ON THE CUBE FACES

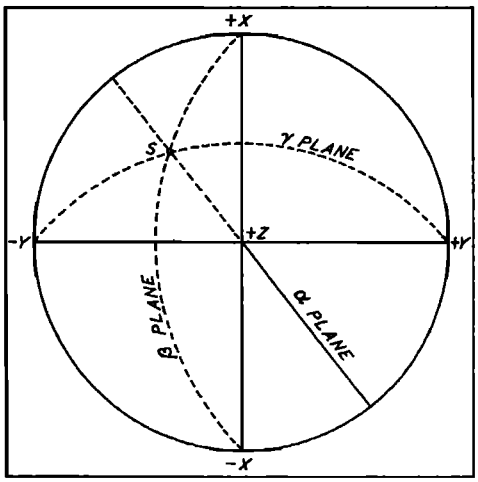


FIG. 31—PROJECTION OF THE S-SEEKING POLE INTO THE LOWER HEMISPHERE $\beta=180°+110°$ $\gamma=180°+40°$

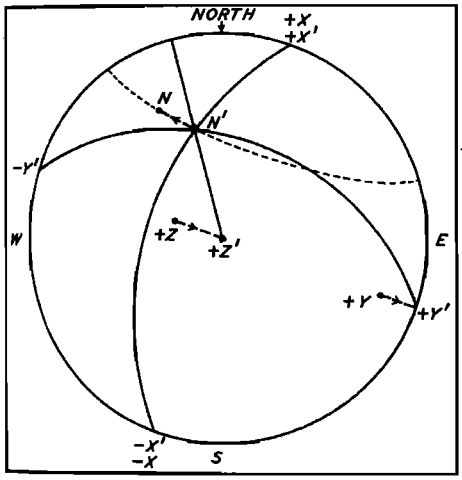


FIG. 32—LOCATION OF TEMPORARY COORDINATES AND ROTATION OF MAGNETIZATION

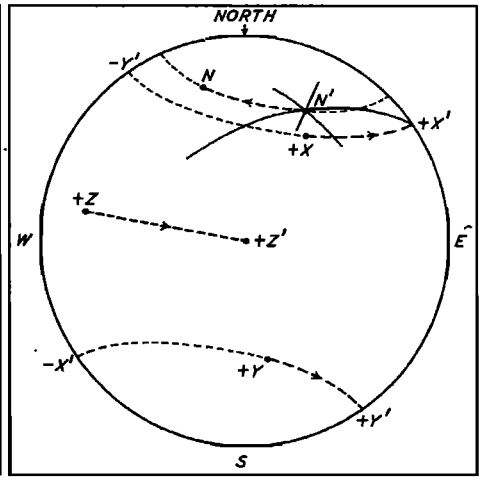


FIG. 33—LOCATION OF TEMPORARY COORDINATES AND ROTATION OF MAGNETIZATION

this common point is the solution for the north-seeking pole of the direction of magnetization. If, in the present case, $+X$ had been toward true north and $+Y$ toward the east, the declination in the sample would be 45° and the inclination 63°. The latter is determined by rotating the point of intersection over the net

so that it lies on one of the calibrated diameters and reading in from the circumference the number of degrees from the horizontal plane.

When it is found that the north-seeking side of the sample does not fall in the lower hemisphere, the determination of the direction of magnetization in a cube requires more care in the plotting. In this case, it is necessary to visualize the south-seeking sides of the planes tilting into the lower hemisphere as the north sides tilt upwards. Consider the example of Figure 31. The north-seeking side of the α plane, having a value of 130°, is plotted as a heavy line, and the south-seeking side a dotted line. As the value of β (290°) is greater than 180° in measuring from $+Y$ through $+Z$, the north-seeking side of the β plane leaves the lower hemisphere at $-Y$ and the south-seeking side appears at $+Y$. Accordingly, the south-seeking side of the β plane is 110° from $+Y$ through $+Z$. The same procedure is followed with γ.

Rotation of coordinate axes—It is not always convenient to cut cubes from oriented specimens in such a way that $+X$, $+Y$, and $+Z$ are directed north, east, and downward. The simplest deviation from this ideal is when $+Z$ is downward and $+X$ and $+Y$ have a known angular relationship to north. In this case, it is most convenient first to indicate the north on the plotting sheet and then measure from it, in the proper sense, the deviations of $+X$ and $+Y$. Then relative to these plotted positions of $+X$, $+Y$, and $+Z$, the values of α, β, and γ are plotted. The relationship of the direction of magnetization to the indicated north may then be read directly.

There are occasions, particularly in work with tilted beds, when it becomes impracticable to cut the cube with its edges in the ideal orientation or even with $+Z$ vertical. However, a procedure is available which makes it possible under these circumstances to determine the polarization at the outcrop with only one additional plotting operation. A flat surface on the bed to be sampled is selected. On it a horizontal line (strike of the bed) is drawn and its bearing determined. An arrow on the line indicates the end nearest north, and is defined as the $+X$ direction. $+Z$ is defined as perpendicular to $+X$ and directed downward perpendicular to the orienting surface on the bed. This inclination is determined. The cube is then cut with its edges parallel to these coordinate axes, and α, β, and γ are measured relative to them.

The orientations in space of the coordinate axes can be plotted directly on the net, it being considered that the projection plane is horizontal. It is inconvenient, however, to plot the magnetic data relative to these coordinate axes when they are not located on the periphery and in the center of the circle of projection. Accordingly, the problem resolves itself into one of plotting relative to new coordinate axes that are in a location making plotting easy, and then rotating these temporary coordinates into their proper orientation in space. The determined direction of magnetization is rotated in the same manner as was necessary to put the temporary coordinates into coincidence with the actual coordinates. This procedure is illustrated in Figure 32. In the selection of the position of the temporary coordinates, it is most convenient to make $+X$ and $+X'$ coincide. Rotation of the temporary coordinates into coincidence with the actual coordinates and of the direction of magnetization N' to its actual position in space relative to the horizon and north is now carried out. The small circles on the net indicate the change in position of

projected directions which are rotated about an axis parallel to the plane of projection. In the present case, $+X - X$ is the rotation axis (the only axis by which it is possible to rotate Y' to Y and Z' to Z, and to retain X' and X in coincidence). The amount of rotation necessary is determined by the angular distance of $+Y'$ from $+Y$. N' is thus rotated 20° to its final position N. N represents the direction in space of the north-seeking end of the direction of magnetization.

Although the field orienting procedure is more elaborate, it is possible, using an extension of the above technique, to determine the magnetization at the outcrop when all three coordinate axes are removed from ideal position. $+X$, $+Y$, and $+Z$ are plotted on the net in the positions they occupied relative to north and to the horizon at the outcrop. The positions of the temporary coordinates for use in plotting must now be selected. Of the many permissible locations, there is one which makes the later necessary rotation a one-step operation. $+Z'$ is placed at the center of the net (Fig. 33) and, with the line joining $+Z + Z'$ placed along the diametral "small circle," the positions of $+X'$ and $+Y'$ are determined by following along the small circles from $+X$ and $+Y$ to the periphery. Plotting of the magnetic data is carried out relative to these temporary axes. The temporary solution of the direction of magnetization N' is then rotated back along one of the family of small circles used in the determination of the temporary axes in the amount that $+ Z'$ is distant from $+Z$. Following this rotation, the direction of magnetization relative to north and to the horizon may be read directly from the plot.